

imparting computer courses to visually-challenged people are functional across India. Resource centres at colleges, like Xavier's Resource Centre for the Visually Challenged (XRCVC) at Mumbai, managed by Dr Sam Taraporewala, imparts computer training to the visually impaired using various technological aids and state-of-the-art computers.

"Along with screen readers, screen magnifiers, and book players, we use other hardware-based aids like Mountbatten MB Pro (a talking / typing Braille), SARA (Scanning and Reading Appliance), Prisma (a CCTV-based device for low-vision users) and Zoom Ex (an instant photo scanner)," says Prashant Naik, Training and Development Officer, XRCVC. The Centre also serves as a test-bench for upcoming Web sites to check for accessibility features and software meant to help the visually-challenged cope with technology.

At these junctions, computer training courses are offered to make the visually-challenged conversant with basic packages like Microsoft Office and with the Internet. "Besides computer-centric training, we at Enable India provide training programmes that involve the development of communication skills in English, analytical skills, and error-proofing techniques to help the visually-challenged perform well even in difficult situations—that is, those where proper eyesight would be considered a must," says Raghavan.

We're now seeing several instances where the visually-challenged are employed for even challenging tasks. The IT industry has expressed tremendous interest in employing such individuals through NGOs like Enable India, as part of their social responsibility. Enable India provides, for the visually-challenged, workplace solutions involving job identification, system configuration, and troubleshooting.

The Next Step

Three major areas where development is taking place on a major scale in India in terms of software are:

- Indian-sounding text-to-speech synthesiser engines for screen readers,
- Tactile imaging embossers (embossers that convert images to embossed dots), and
- Screen readers for different browsers.

The Internet being the ocean of knowledge that it is, a majority of the developments in existing and upcoming software is taking place for better Web accessibility.

Loads of screen readers, as well as tools like Opera's voice-commands feature are available, but why aren't Indian citizens being made aware of them? Well, there is the accent problem with us Indians, and not every Web page is designed with a screen reader in mind.

Web pages scripted with different languages—for instance, AJAX carrying dynamic JavaScript—make it difficult for screen reader software like JAWS to cope. To combat the browser issue, Mane, with other global developers, is working on the Accessibility functions in the Firefox browser. He believes that Web

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pages should be formatted using W3C-certified scripting languages.

The cheaper and affordable Indian-version screen reader called SAFA (Screen Access For All) is under development, led by Dipendra Manocha, director for IT and Services at the National Association for the Blind (NAB). Manocha also works as assistant project manager for DAISY for All-India, a project of the DAISY consortium; DAISY (Digital Accessible Information System) is an open standard used for developing tools such as digital talking books.

At XRCVC, e-Signs (banking software for the visually-challenged) is being tested and developed by CMC Ltd; it was conceptualised by the Centre. This software will help banks with thumb-print recognition reach an expected accuracy closer to 100 per cent for thumb-prints by the visually-challenged. Besides, the Centre recently tested an open source screen reader called AccessibilityWorks, developed by IBM, which works with Firefox on Linux.

On the open source front, at LinuxAsia 2007, Klaus Knopper with his wife Adriane (who is visu-



Krishnakant Mane talks to software developers and students, helping them understand what the visually challenged need from software

ally-impaired) announced the development of Adriane Knoppix, a free operating system (GNU/Linux) specially for the visually-challenged.

An IT student of Agra College recently developed E-Netra, a device using which the visually-challenged will hear the text they can't read; it's a gadget with a zoom lens and unique software.

Go Open Source

Mane reminds us that almost 80 per cent of Web servers run on GNU/Linux-based operating systems. IT giants and many big multinationals now use GNU/Linux-based OSes for their servers.

Now, though GNU/Linux-based OSes and programs are free, why is the common man still unaware of their benefits? Mane laments, "Free-of-cost programs are very costly, as the willingness to share is rare. If the government and people gain some awareness soon, they'll realise the benefits of viable and affordable open source systems over the costly proprietary ones that lack Indian speech synthesis and language support."

Tools Currently Used In India

Software and hardware aids widely used by various NGOs and Resource Centres

SCREEN READERS
JAWS: Job Access With Speech is a proprietary screen reading software that converts text to speech and reads it out at an adjustable place, with options for voice (male/female), tone, languages, and speech engines.
www.freedomscientific.com/fs_products/software_jaws.asp

ORCA: An open source application for the visually challenged with many features, including a speech synthesiser; it supports Braille and also a text magnifier.
<http://live.gnome.org/Orca>

Emacspeak: An open source text-to-speech interface meant for visually-challenged users as an enabler to browse the Web and use various programs without assistance.
<http://emacspeak.sourceforge.net>

SAFA (Screen Access For All): A cheaper alternative to the JAWS screen reader, which uses an Indian-sounding synthesiser for the speech engine.
<http://safa.sourceforge.net>

MOBILE PHONES
Mobilespeak: A screen reader application with Braille support for

Symbian Series 60 Edition cell phones.
www.codefactory.es/en/products.asp?id=24

TALKS: A screen reader that supports 20 languages on Symbian Series 60 phones.
www.nuance.com/talks

Call History: Software that allows visually-impaired users to hear a description of dialled, missed, and received calls, along with voice-prompt-based navigation.
www.tinfomobile.com/Applications.html

SCREEN / TEXT MAGNIFIERS
MAGic: Screen magnification software with hotkeys different from those in Windows.
www.freedomscientific.com/fs_products/software_magic.asp

ZoomText: A text magnifier and reading software for low-vision users.
www.aisquared.com/index.cfm

TYPING
Talking Typing Teacher: Meant for novices as well as advanced users, this is a typing tutor with speech used for every function
www.braillebookstore.com/talking-typing-teacher.htm

Braille: Duxbury's Braille Translator

Software that provides translation, formatting, and word-processing facilities to automate the process of working from regular text to Braille and vice-versa.
www.duxburysystems.com/dbt.asp

WinBraille: A Braille editor meant for Index Braille embossers.
www.indexbrailleaccessibility.com/downloads/winbraille.htm

Shree-Lipi Braille (Indian-language Braille translation software): Has support for Indian languages including Hindi, Marathi, Sanskrit, Gujarati, and others, with the main function of enabling translation of text to Braille. The file created can be used with a Braille embosser to print out content in Braille.
www.modular-infotech.com/html/braille.html

PLAYBACK SOFTWARE
AMIS: (Adaptive Multimedia Information System) for the DAISY Book Reader
An open source book-reading software for talking books saved in the DAISY format, with support for South Asian languages.

This list represents software widely used in India; it is not exhaustive.



Mane himself is quite comfortable on his laptop

This idea has hit Kerala state. Today, over 80 IT resource centres provide training facilities to people in rural areas as well as to visually-challenged people on computers running open source OSes with the Orca screen reader installed for the benefit of the latter.

Future Expectations

Like J K Rowling wrote in *Harry Potter and the Order of the Phoenix*, "It is not our abilities but our

choices which make us what we are." Visually-challenged people are ready to take up the challenges of life—now with technology at their side. Many tech evangelists have predicted that the future of computing is mobile; usage patterns seem to indicate the truth of this.

Neha Trivedi, Project Officer, XRCVC, says, "In the near future, a multifunction portable gadget / device that can do nearly most functions will increase the flexibility of performing tasks like text-reading, scanning, and the use of speech technology almost anywhere. This could make the visually-challenged more independent."

Various aids, including software, audio players, and multifunctional gadgets do exist for the visually-challenged. But they're expensive for the most part, and quite unaffordable for most of those who need them.

You can help. Donate old working PCs to IT resource centres run by your state government, or trusts for the visually challenged, or at NGOs like Enable India and many more (take a look at the *Touched by Tech* section in our July 2007 issue). At the Talking Libraries and resource centres—like XRCVC in Mumbai—you can volunteer to scan printed material for conversion to audio-books. A lot of people will be happy. ■

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